

SpeakerScope

Live speaker-attributed transcription

who said what, as they say it

AssemblyAI x lablab.ai Hackathon | September 2026

AssemblyAI Universal Streaming + NVIDIA Nemotron-3-Diarization (on-device)



The Problem

Transcripts know what was said - not who said it

Flat transcripts

No speaker labels: meetings become a wall of text - unusable for recaps, QA, or action items.

Cloud diarization

Post-hoc upload, minutes of latency, per-minute billing, and raw audio leaves the device.

Real-time need

Live meeting assist, call coaching, accessibility: labels must appear while people speak.

Gap in the market: fast, private, speaker-attributed transcription running on commodity desktop hardware.

The Solution

Two best-in-class engines fused in one native C++ app

WHAT was said

AssemblyAI Universal Streaming

- Real-time WebSocket (v3), 16 kHz PCM
- Word-level timestamps, sub-second partials
- Clean session lifecycle - billing-safe

WHO is speaking

NVIDIA Nemotron-3-Diarization

- 100M-param model on-device (GGUF q8_0)
- Native C++ via audio.cpp - no GPU, no Python
- 8 channels, overlap-aware, streaming

Timestamp-domain fusion -> every word gets a speaker label live, shown as Person 1..8 lanes users can rename, filter and export.

Architecture

One capture, two consumers, one fused transcript

AUDIO mic / WASAPI loopback / WAV -> 16 kHz mono lock-free ring

v

STT AssemblyAI v3 WS -> 50 ms pcm_s16le -> turns + word timestamps

v

DIAR audio.cpp Nemotron-3 -> streaming inference -> open speaker turns

v

FUSE word-midpoint argmax + hysteresis + overlap flag

v

UI Dear ImGui - live lanes, rename people, filter/copy/export per person

Engineering highlights

Open-turn streaming patch

audio.cpp decoder patched to emit live open turns - speakers labeled while talking, not only after pauses.

7.6x realtime on CPU

Custom chunk geometry (chunk_len=40) vs. 0.53x for the stock 'low' profile - identical accuracy, ~5 s label granularity.

Tee architecture

STT and diarization consume separate rings - local inference can never starve the realtime AssemblyAI stream.

Beyond 8 speakers

8-channel limit is concurrency-only; voice-embedding clustering resolves unlimited global Person identities.

Privacy

Diarization fully on-device; only transcription audio goes to AssemblyAI.

Verified results

It runs - headless and in the GUI

117/117

automated tests green - incl. 80-
case generated-meeting battery

55.8 s

4-voice wav -> live Nemotron +
AssemblyAI -> attributed
transcript

7.6x RT

on-device diarization
throughput on a 12-core CPU

<1 s

session stop - STT socket closed
before diar drain (billing-safe)

Demo artifacts

- speakerscope.exe - GUI: live lanes, rename Person N, filter & export per person
- speakerscope_sim.exe - headless end-to-end: wav -> real AssemblyAI + Nemotron -> fused transcript
- speakerscope_diar_bench.exe - model throughput benchmark (16x RT max profile)
- Sample output: '[Person 2] The diarization model now supports up to 8 simultaneous speakers.'

Business value & stack

A transcript tells you what was said. A record tells you who said it.

- Hybrid teams - searchable, attributable meeting recaps
- Call centers - agent/customer separation for QA and coaching
- Accessibility - follow conversations by speaker in real time
- Privacy-sensitive settings - diarization never leaves the device

Stack

C++20 | CMake + vcpkg | Dear ImGui + GLFW | miniaudio | IXWebSocket + mbedTLS | AssemblyAI Streaming v3 | audio.cpp (GGUF) | Nemotron-3-Diarization q8_0 | GoogleTest | Windows, MIT license

SpeakerScope - live, on-device, speaker-attributed transcription.